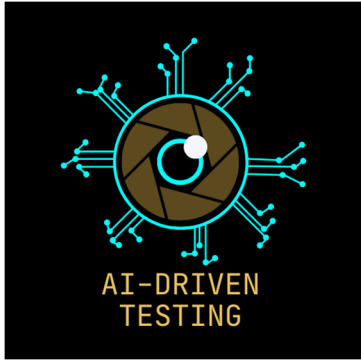
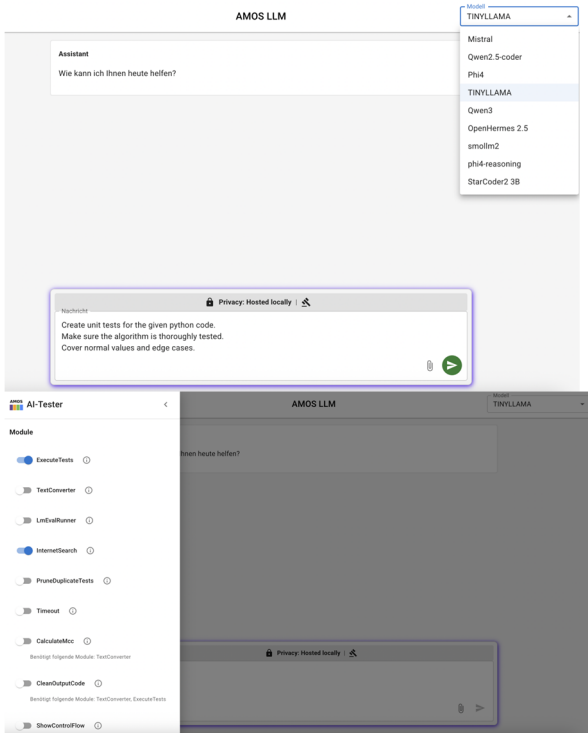


AMOS SS25 Project 4 ai driven testing

Project name	AI Driven Testing
Project mission	Deliver a minimum viable product (MVP) of an AI-powered test generation assistant that interacts effectively with real-world software projects. This includes laying the empirical and research groundwork for the idea. From this research we will conclude a MVP, that can analyze simple Python and C++ programs and produce corresponding test cases. Operated through a user-friendly chat interface. The MVP will showcase initial integration with widely used open-source testing frameworks such as Robot Framework, proving the viability of AI-assisted test automation in practical, everyday development environments.
Industry partner	GRAU DATA GmbH
Team logo	 The logo features a stylized camera aperture in the center, with circuit-like lines radiating outwards. Below the aperture, the text "AI-DRIVEN TESTING" is written in a bold, sans-serif font.
Project summary	We had the great opportunity to get hands on experience partaking in AI driven agile software project developing an LLM based solution for generating tests. Furthermore we incorporated university grade research to validate our implementation using industry standard methods. Our core finding is that the quality of the LLM Response is not necessarily correlated to the parameter size. Beyond that our essential find lays in the comprehensive wiki documentation https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki .
Project illustration	 The screenshot shows the AMOS LLM web interface. At the top, there's a header with "AMOS LLM" and a dropdown menu for selecting the model, currently set to "TINYLLAMA". Below the header is a chat area with a message from the "Assistant" asking "Wie kann ich Ihnen heute helfen?". A modal window is open in the center, displaying a task: "Create unit tests for the given python code. Make sure the algorithm is thoroughly tested. Cover normal values and edge cases." At the bottom, there's a sidebar menu for the "AI-Tester" application, listing modules like ExecuteTests, TextConverter, LintRunner, InternetSearch, PruneDuplicateTests, Timeout, CalculateMcc, CleanOutputCode, and ShowControlFlow.
Team photo	
Project repository	https://github.com/amosproj/amos2025ss04-ai-driven-testing
Additional information	